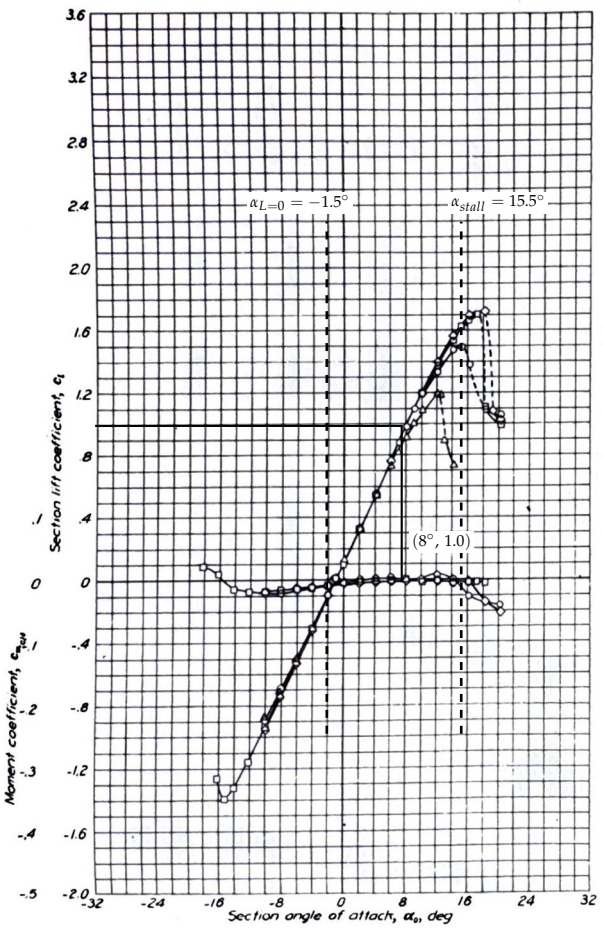
Worked Solutions

Wing aerodynamics and geometry

Question 1 — NACA 23015: lift-curve slopes and stall speed (Exam 2025–26, Q2(c))



From Figure 1 ($\circ$ symbols, $Re = 2.6 \times 10^6$):

- Stall angle: $\alpha_{stall} \approx \mathbf{15.5}^\circ$ [2]
- Zero-lift angle: $\alpha_{L=0} \approx -\mathbf{1.5}^\circ$ [2]
- Linear range: $c_l \approx 1.0$ at $\alpha = 8^\circ$

2D lift-curve slope [2]

$$a_0 = \frac{\Delta c_l}{\Delta \alpha} = \frac{1.0 - 0}{8 - (-1.5)} = 0.105 \text{ /deg} = 6.02 \text{ /rad}$$

3D lift-curve slope ($A = 8$, $e = 1$) [2]

$$a = \frac{a_0}{1 + \dfrac{a_0}{\pi e A}} = \frac{6.02}{1 + \dfrac{6.02}{\pi \times 1 \times 8}} = \frac{6.02}{1.24} = 4.86 \text{ /rad} = 0.085 \text{ /deg}$$

Stall speed (steady level flight, $L = W$) [4]

$$W = mg = 950 \times 9.8 = 9{,}310 \text{ N}$$

$$C_{L,max} = a\,(\alpha_{stall} - \alpha_{L=0}) = 0.085 \times (15.5 + 1.5) = 0.085 \times 17 = 1.45$$

$$W = \tfrac{1}{2}\rho V_{stall}^2 S\, C_{L,max} \;\Rightarrow\; V_{stall} = \sqrt{\frac{2W}{\rho\, S\, C_{L,max}}} = \sqrt{\frac{2 \times 9{,}310}{1.225 \times 16 \times 1.45}} = \mathbf{25.6\ m/s}$$

$\alpha_{stall} \approx 15.5°$, $\alpha_{L=0} \approx -1.5°$, $a_0 = 0.105$/deg (6.02/rad), $a = 0.085$/deg (4.86/rad), $C_{L,max} \approx 1.45$, $V_{stall} \approx 25.6$ m/s

Points to stress: the 3D slope is lower than the 2D slope because of the induced angle of attack ($\alpha_i = C_L/\pi Ae$), so the wing needs a higher angle of attack for the same lift. This method assumes the wing stalls at the aerofoil stall angle. Real wings reach a lower $C_{L,max}$ (roughly $0.9\,c_{l,max}$), so the real stall speed would be slightly higher.

Question 2 — Lift coefficient at the beginning and end of cruise

Wing area. From the assumed take-off wing loading (S-3 Viking value):

$$S = \frac{W_0}{W_0/S} = \frac{56{,}706}{88} = 644 \text{ ft}^2$$

Dynamic pressure at $M\,0.6$, 30,000 ft:

$$q = \tfrac{1}{2}\rho V^2 = \tfrac{1}{2}\,(8.907 \times 10^{-4})\,(596.9)^2 = 158.7 \text{ lb/ft}^2$$

$$\big(\text{check: } q = 0.7\,p\,M^2 = 0.7 \times 629.7 \times 0.6^2 = 158.7 \text{ lb/ft}^2\big)$$

Weights.

$$W_2 = W_0\,\frac{W_1}{W_0}\,\frac{W_2}{W_1} = 56{,}706 \times 0.970 \times 0.985 = 54{,}180 \text{ lb} \qquad W_3 = 0.8581\,W_2 = 46{,}490 \text{ lb}$$

Lift coefficient ($L = W$ in level cruise): $\quad C_L = \dfrac{W}{q\,S}$

$$C_{L,\,start} = \frac{54{,}180}{158.7 \times 644} = \mathbf{0.53} \qquad C_{L,\,end} = \frac{46{,}490}{158.7 \times 644} = \mathbf{0.455}$$

Beginning of cruise: $C_L \approx 0.53$	End of cruise: $C_L \approx 0.45$	(a 14% drop)

For the return cruise (segment 5), the same steps give $W_4 = 43{,}130$ lb $\rightarrow C_L = 0.42$ and $W_5 = 37{,}010$ lb $\rightarrow C_L = 0.36$.

Discussion.

- As fuel burns, W falls but q is fixed (constant M and altitude), so C_L falls and the aircraft moves away from its best L/D. Holding C_L (and L/D) constant requires a *cruise-climb*: $W/(\rho S)$ constant, so the aircraft climbs as it gets lighter. In practice this is approximated with step climbs.
- Going further (with $A = 8$, $e = 0.8$ and $(L/D)_{max} = 16$): $C_{D0} = \pi Ae/[4(L/D)^2_{max}] = 0.0196$, and $(L/D)_{max}$ occurs at $C_L^* = \sqrt{\pi Ae C_{D0}} = 0.63$. At $C_L = 0.53$, $L/D \approx 15.8$, and at $C_L = 0.455$, $L/D \approx 15.2$. The $0.866\,(L/D)_{max} = 13.9$ assumed in sizing is therefore conservative for this wing loading.

Question 3 — Wing planform selection

Starting point, the S-3 Viking: $A = b^2/S = 68.7^2/598 = 7.9$; $W/S = 52{,}539/598 = 88\,\text{lb/ft}^2$; $\Lambda_{LE} \approx 15°$; t/c = 16% root, 12% tip; high wing.

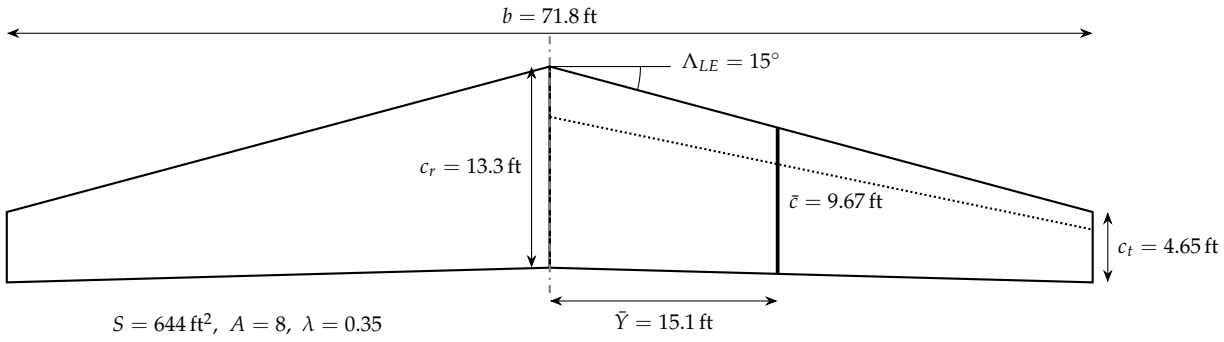
Parameter	Selected	Reasoning
Aspect ratio, A	**8**	Trend for military cargo/bomber: $A = 5.57\,(0.75)^{-1.075} = 7.6$. The S-3 has 7.9. The long (3 hr) loiter rewards low induced drag, so lean high: $A = 8$.
Sweep, Λ_{LE}	**15°**	Cruise M 0.6 is below the drag rise, so only a small sweep is needed (as on the S-3). It gives some M_{cr} margin for the $M \approx 0.75$ dash and some help with c.g. placement. $\Lambda_{c/4} \approx 11.7°$.
Taper ratio, λ	**0.35**	Low-sweep wing: $\lambda \approx$ 0.4–0.45 gives near-elliptic loading; the modest sweep moves load outboard, so slightly more taper is used. A near-straight trailing edge (as on the S-3) eases flap and spar layout.
Thickness, t/c	**16% root, 12% tip**	Low M allows a thick wing: lighter structure, large fuel volume (fuel fraction 0.38), high $C_{L,max}$ and a gradual stall.
Wing position	**High**	Ground clearance for underwing engines, the weapons bay and sonobuoy launch tubes; matches the S-3.
Dihedral	**0° to 2°**	Table for a high, nearly unswept wing: 0 to 2°. The high wing already adds roll stability. Select $\approx 1°$.
Twist	$-3°$ **wash-out**	Tip nose-down so the root stalls first and the ailerons stay effective.
Incidence	$\approx 0°$–$1°$	Military/transport guideline; set so the fuselage is at minimum-drag attitude at the cruise C_L.

Resulting geometry ($S = 644\,\text{ft}^2$, $A = 8$, $\lambda = 0.35$):

$$b = \sqrt{A\,S} = \sqrt{8 \times 644} = 71.8\ \text{ft}, \qquad c_r = \frac{2S}{b(1+\lambda)} = \frac{2 \times 644}{71.8 \times 1.35} = 13.3\ \text{ft}, \qquad c_t = \lambda c_r = 4.65\ \text{ft}$$

$$\bar{c} = \frac{2}{3}\,c_r\,\frac{1+\lambda+\lambda^2}{1+\lambda} = \frac{2}{3}(13.3)\frac{1.4725}{1.35} = 9.67\ \text{ft}, \qquad \bar{Y} = \frac{b}{6}\,\frac{1+2\lambda}{1+\lambda} = \frac{71.8}{6}\,\frac{1.70}{1.35} = 15.1\ \text{ft}$$

Quarter-chord sweep: $\tan\Lambda_{c/4} = \tan\Lambda_{LE} - \dfrac{1-\lambda}{A(1+\lambda)} = 0.268 - 0.060 = 0.208 \Rightarrow \Lambda_{c/4} = 11.7°$.



$A = 8,\ \lambda = 0.35,\ \Lambda_{LE} = 15°,\ t/c = 0.16/0.12,$ high wing, $\Gamma \approx 1°,$ twist $-3°$
$b = 71.8\,\text{ft},\ c_r = 13.3\,\text{ft},\ c_t = 4.65\,\text{ft},\ \bar{c} = 9.67\,\text{ft}$ at $\bar{Y} = 15.1\,\text{ft}$

Other reasonable answers: A = 7.5–9, λ = 0.3–0.45 and Λ_{LE} = 0–20°, provided the choice is justified against the mission and the S-3.

Question 4 — Why tails are not scaled-down copies of the wing

A wing is designed to carry a lot of lift efficiently. A tail is designed to provide **trim, stability and control**, and it normally works at only a fraction of its lift capability. Its planform is chosen for that job:

1. **The tail must stall after the wing.** A lower aspect ratio raises the stall angle of attack, because the tip vortices reduce the effective angle over more of the span. Horizontal tails therefore use $A \approx 3$–5, against 7–9 for a transport/ASW wing, so elevator authority is kept when the wing is near or past stall.
2. **More sweep and less thickness.** The horizontal tail's leading-edge sweep is usually about 5° more than the wing's, and on high-speed aircraft its t/c is about 10% thinner. Both raise the tail's M_{cr}, so shocks form on the wing first and elevator effectiveness is not lost. Extra sweep also delays tail stall and lengthens the vertical-tail moment arm.
3. **Lift in both directions.** Tails must produce up and down loads (or left and right for the fin), so they use symmetric aerofoils. A wing uses cambered sections for lift in one direction.
4. **Induced drag matters less.** Tails carry little lift in cruise, so the induced-drag benefit of a high aspect ratio is small. They are sized by the tail volume coefficient, not by L/D.
5. **Structure and weight.** A short-span, low-A tail has lower root bending moments, is lighter and stiffer, and is less prone to flutter. That matters because it sits on a slender aft fuselage, far from the c.g.
6. **Arrangement effects.** On a T-tail the fin carries the horizontal tail, so it uses a low A (0.7–1.2) and little taper (λ = 0.6–1.0) for stiffness. The endplate effect allows a smaller fin. Some GA aircraft use untapered tails ($\lambda = 1$) to cut manufacturing cost.
7. **Operating environment.** The horizontal tail works in the wing's downwash and wake, and the fin must stay effective at high angles of attack and in spins. These favour robust, low-A surfaces over efficient high-A ones.

In one line: the wing is optimised for lift-to-drag, whereas tails are optimised to *never* stall or meet compressibility before the wing, to lift in both directions, and to be light and stiff at the end of the fuselage. This leads to lower aspect ratio, more sweep, thinner symmetric sections and different taper.